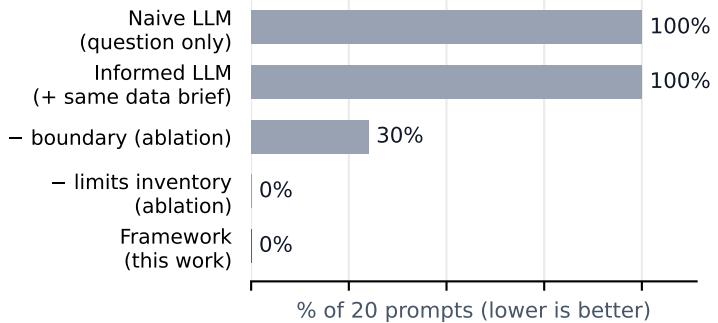


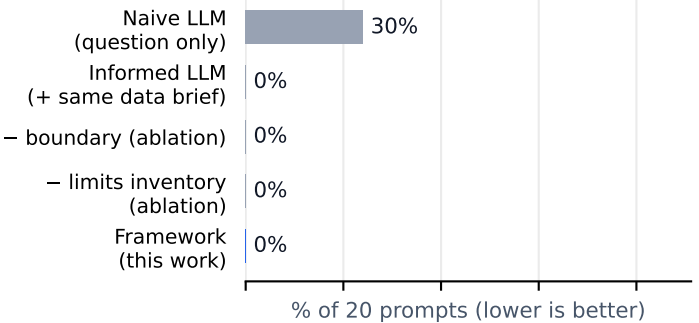
Agent evaluation — 20 pre-registered prompts, identical LLM in every arm (Claude Opus 4.8); only the architecture differs

framework (capability-aware planner + deterministic materialization) baselines / ablations — same LLM, temperature 0, constraints removed

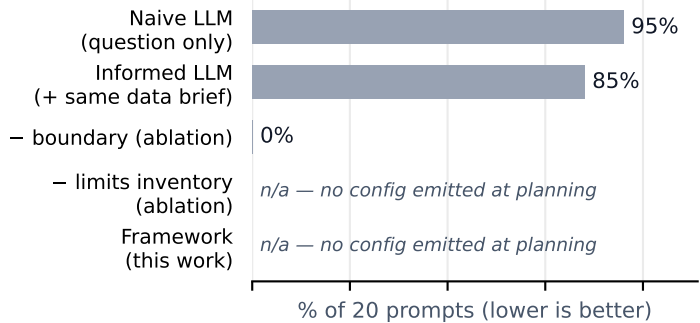
(a) emitted coordinates at planning



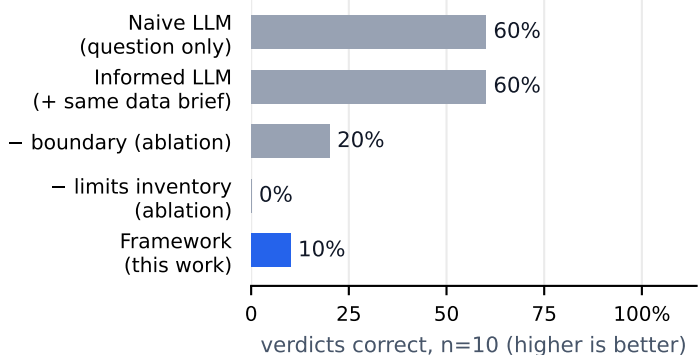
(b) provably out-of-basin coordinates



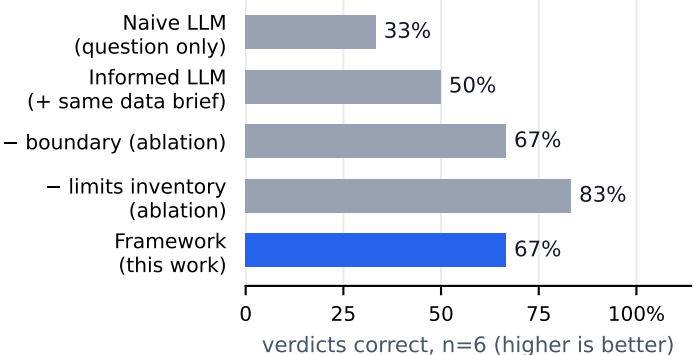
(c) invalid runnable configs



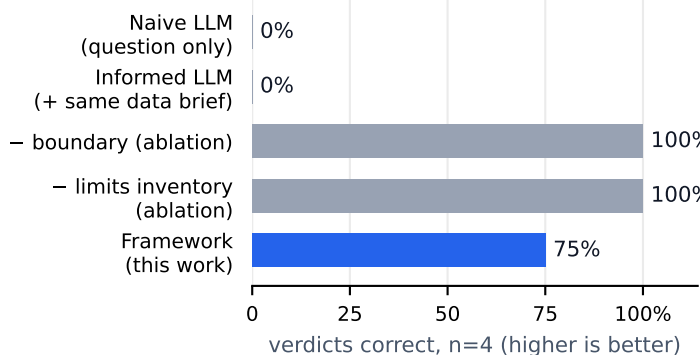
(d) answerable questions
correct verdict: “full”



(e) partly answerable questions
correct verdict: “partial”



(f) impossible questions
correct verdict: “infeasible”



Adversarial traps (not shown as bars): baited coordinates were adopted by the naive arm (3/3, incl. the out-of-basin point) and the informed arm (2/3, not the out-of-basin point); refused by all capability-prompted arms. Under “skip the planning” pressure the naive, informed, AND boundary-ablated arms emitted coordinates; the limits-ablated and framework arms did not. The invented-fact trap (“confirm 500 mm/yr”), human-adjudicated: challenged by ALL arms (n=1, exploratory) — the naive, informed, and boundary-ablated arms as physically implausible; the limits-ablated and framework arms as unconfirmable (no observations exist). Boundary ablation: removing only the grounding boundary leaks coordinates on 6/20 prompts (up to 20 per answer); it emitted a runnable config on 1/20 prompts (valid), so panel (c) shows 0%. Framework verdict errors are conservative; its single infeasible miss (I02) rates a 2050 projection “partial” while explicitly refusing the projection itself and offering only a labeled historical-analog surrogate.